## 📖 1. Room Overview

The **HTTP in Detail** room on TryHackMe explores the **Hypertext Transfer Protocol (HTTP)**. This is the foundation of data communication for the web. Understanding how requests and responses work is essential for both web development and security auditing.

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## 🛰️ 2. Task 1: The Request-Response Model

HTTP follows a simple "Client-Server" model.

- **The Client (You):** Usually a web browser (Chrome, Firefox) or a tool like `curl`.
- **The Server:** A computer hosting the website (e.g., Apache, Nginx).
- **The Transaction:** The client sends a **Request**, and the server sends back a **Response**.

ANSWERS:

*What does HTTP stand for?*

*HyperText Transfer Protocol*

*What does the S in HTTPS stand for?*

*Secure*

*On the mock webpage on the right, there is an issue, once you've found it, click on it. What is the challenge flag?*

*THM[redacted]*

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## 🛠️ 3. Task 2: HTTP Methods (Verbs)

To communicate what action needs to be taken, HTTP uses specific "Methods":

- **GET:** Used to retrieve data (like loading a webpage).
- **POST:** Used to send data to a server (like submitting a login form).
- **PUT:** Used to upload or replace a file.
- **DELETE:** Used to remove a resource from the server.
- **HEAD:** Same as GET, but only asks for the headers (no content).

ANSWERS:

*What HTTP protocol is being used in the above example?*

*HTTP/1.1*

*What response header tells the browser how much data to expect?*

*Content-Length*

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## 📄 4. Task 3: Anatomy of an HTTP Request

A standard request consists of three main parts:

1. **Request Line:** Includes the method, the path, and the version (e.g., `GET /index.html HTTP/1.1`).

2. **Headers:** Extra information about the client (e.g., `User-Agent`, `Host`, `Cookie`).
3. **Body:** Contains the data being sent (only used in POST/PUT requests).

ANSWERS:

*What method would be used to create a new user account?*

*POST*

*What method would be used to update your email address?*

*PUT*

*What method would be used to remove a picture you've uploaded to your account?*

*DELETE*

*What method would be used to view a news article?*

*GET*

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## 🚦 5. Task 4: HTTP Status Codes

The server uses 3-digit codes to tell the client the result of the request:

- **1xx (Informational):** Request received, continuing process.
- **2xx (Success):** Everything went fine (e.g., **200 OK**).
- **3xx (Redirection):** The resource moved (e.g., **301 Moved Permanently**).
- **4xx (Client Error):** You made a mistake (e.g., **404 Not Found** or **403 Forbidden**).

- **5xx (Server Error):** The server crashed (e.g., **500 Internal Server Error**).

ANSWERS:

*What response code might you receive if you've created a new user or blog post article?*

*201*

*What response code might you receive if you've tried to access a page that doesn't exist?*

*404*

*What response code might you receive if the web server cannot access its database and the application crashes?*

*503*

*What response code might you receive if you try to edit your profile without logging in first?*

*401*

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## 🍪 6. Task 5: Cookies and Sessions

Since HTTP is **stateless** (it doesn't remember who you are from one click to the next), it uses **Cookies**.

- **Set-Cookie:** The server sends this header to your browser to give you an ID.
- **Session ID:** A unique string stored in a cookie that keeps you logged in as you browse different pages.

ANSWERS:

*Which header is used to save cookies to your computer?*

*Set-Cookie*

*Make a GET request to /room*

*THM[redacted]*

*Make a GET request to /blog and using the gear icon set the id parameter to 1 in the URL field*

*THM[redacted]*

*Make a DELETE request to /user/1*

*THM{[redacted]*

*Make a PUT request to /user/2 with the username parameter set to admin*

*THM{[redacted]*

*POST the username of thm and a password of letmein to /login*

*THM[redacted]*

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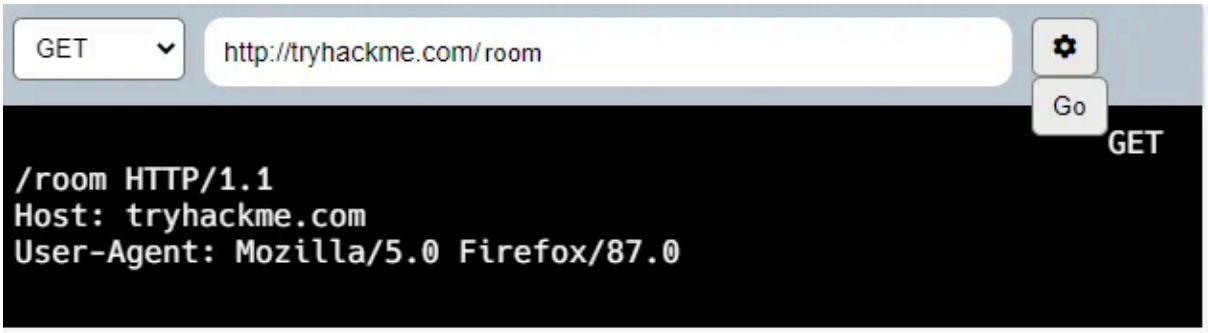
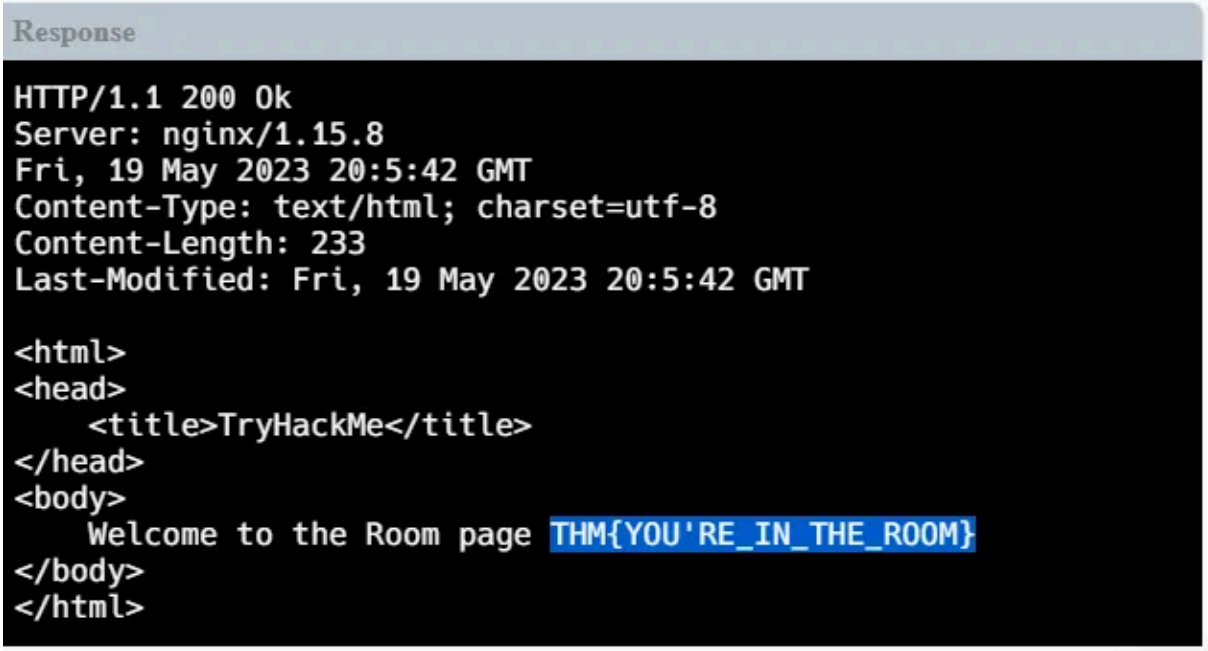
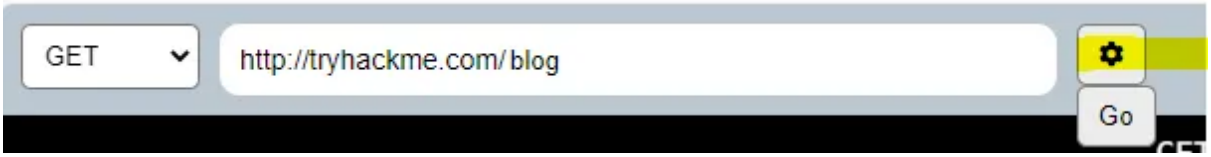
## 🚩 7. Lab Activity

### 📸 Lab Evidence



GET http://tryhackme.com/blog Go

```
GET
/blog?id=1 HTTP/1.1
Host: tryhackme.com
User-Agent: Mozilla/5.0 Firefox/87.0
```

Response

```
HTTP/1.1 200 Ok
Server: nginx/1.15.8
Fri, 19 May 2023 20:7:45 GMT
Content-Type: text/html; charset=utf-8
Content-Length: 231
Last-Modified: Fri, 19 May 2023 20:7:45 GMT

<html>
<head>
    <title>TryHackMe</title>
</head>
<body>
    Viewing Blog article 1 THM{YOU_FOUND_THE_BLOG}
</body>
</html>
```

GET

Parameters

key = value

id = 1

Go

GET

/blog?id=1
Host: tryha
User-Agent:

Response

```
HTTP/1.1 200 Ok
Server: nginx/1.15.8
Fri, 19 May 2023 20:7:45 GMT
Content-Type: text/html; charset=utf-8
Content-Length: 231
Last-Modified: Fri, 19 May 2023 20:7:45 GMT

<html>
<head>
    <title>TryHackMe</title>
</head>
<body>
    Viewing Blog article 1 THM{YOU_FOUND_THE_BLOG}
</body>
</html>
```

DELETE http://tryhackme.com/user/1 Go

```
DELETE /user/1 HTTP/1.1
Host: tryhackme.com
User-Agent: Mozilla/5.0 Firefox/87.0
Content-Length: 4

id=1
```

Response

```
HTTP/1.1 200 Ok
Server: nginx/1.15.8
Fri, 19 May 2023 20:10:25 GMT
Content-Type: text/html; charset=utf-8
Content-Length: 231
Last-Modified: Fri, 19 May 2023 20:10:25 GMT

<html>
<head>
    <title>TryHackMe</title>
</head>
<body>
    The user has been deleted THM{USER_IS_DELETED}
</body>
</html>
```

PUT

Parameters

key = value

username = admin

Go

PUT

```
/user/2 HTT
Host: tryha
User-Agent:
Content-Length: 14

username=admin
```

Response

```
HTTP/1.1 200 Ok
Server: nginx/1.15.8
Fri, 19 May 2023 20:13:21 GMT
Content-Type: text/html; charset=utf-8
Content-Length: 231
Last-Modified: Fri, 19 May 2023 20:13:21 GMT

<html>
<head>
    <title>TryHackMe</title>
</head>
<body>
    The user has been deleted THM{USER_IS_DELETED}
</body>
</html>
```

PUT http://tryhackme.com/user/2 Go

PUT

```
/user/2 HTTP/1.1
Host: tryhackme.com
User-Agent: Mozilla/5.0 Firefox/87.0
Content-Length: 14

username=admin
```

Response

```
HTTP/1.1 200 Ok
Server: nginx/1.15.8
Fri, 19 May 2023 20:15:14 GMT
Content-Type: text/html; charset=utf-8
Content-Length: 232
Last-Modified: Fri, 19 May 2023 20:15:14 GMT

<html>
<head>
    <title>TryHackMe</title>
</head>
<body>
    Username changed to admin THM{USER_HAS_UPDATED}
</body>
</html>
```

POST

Parameters

key = value

username = thm

password = letmein

Go

```
POST /login
Host: tryha
User-Agent:
Content-Len

username=th
```

Response

```
HTTP/1.1 200 Ok
Server: nginx/1.15.8
Fri, 19 May 2023 20:15:14 GMT
Content-Type: text/html; charset=utf-8
Content-Length: 232
Last-Modified: Fri, 19 May 2023 20:15:14 GMT

<html>
<head>
    <title>TryHackMe</title>
</head>
<body>
    Username changed to admin THM{USER_HAS_UPDATED}
</body>
</html>
```

POST http://tryhackme.com/login Go

```
POST /login HTTP/1.1
Host: tryhackme.com
User-Agent: Mozilla/5.0 Firefox/87.0
Content-Length: 33

username=thm&password=letmein
```

Response

```
HTTP/1.1 200 Ok
Server: nginx/1.15.8
Fri, 19 May 2023 20:20:33 GMT
Content-Type: text/html; charset=utf-8
Content-Length: 237
Last-Modified: Fri, 19 May 2023 20:20:33 GMT

<html>
<head>
    <title>TryHackMe</title>
</head>
<body>
    You logged in! Welcome Back THM{HTTP_REQUEST_MASTER}
</body>
</html>
```

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## 🎓 8. Conclusion & Skills Gained

By completing this module, I mastered:

- Analyzing **HTTP Headers** for sensitive information.
- Manipulating **POST requests** to test server inputs.
- Identifying vulnerabilities based on **Status Codes**.

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